



EmotionCam User Manual

Local visible-expression estimates with privacy-first personalized calibration

Version 1.0.0 | Updated June 16, 2026



EmotionCam 1.0.0

Local visible-expression estimates with privacy-first personalized calibration

Updated June 16, 2026

EmotionCam estimates visible facial expressions. It does not know or guarantee a person's true emotions.

1. Introduction

EmotionCam is a Windows desktop app for one person using a laptop webcam. It shows a local camera preview, estimates visible facial expressions, smooths the result, and displays a label, group, confidence estimate, safe interaction message, and recent timeline.

EmotionCam can also be personalized through local calibration, run in the Windows tray, save metadata-only logs, show local Statistics, and optionally send daily text summaries when explicitly enabled.

2. Privacy and Safety

- Webcam frames are processed locally by default.
- EmotionCam does not identify people or perform face matching.
- There is no account system, telemetry, analytics, automatic screenshot, or video-recording feature.
- Metadata logs contain expression labels/groups, confidence, status, messages,

popup status, and FPS.

- Calibration stores local normalized feature data and labels.
- Raw calibration frames are not saved unless the explicit debugging option is enabled.
- Daily email summaries are off by default and send text only if enabled.

Do not use EmotionCam for medical, psychological, legal, hiring, security, surveillance, or lie-detection decisions.

3. Installation

Install the packaged app

- 1 Open the repository Releases page.
- 1 Download EmotionCam_Setup.exe from the latest release.
- 1 Run the installer.
- 1 Launch EmotionCam from the Start Menu or optional desktop shortcut.
- 1 Allow Windows camera access for desktop apps if prompted.

The app installs per user under:

%LOCALAPPDATA%\Programs\EmotionCam

The installer is unsigned, so Windows SmartScreen may show a warning. Choose

More info > Run anyway only if you trust the downloaded file.

Run from source

```
py -3.12 -m venv .venv
```

```
.\.venv\Scripts\Activate.ps1
```

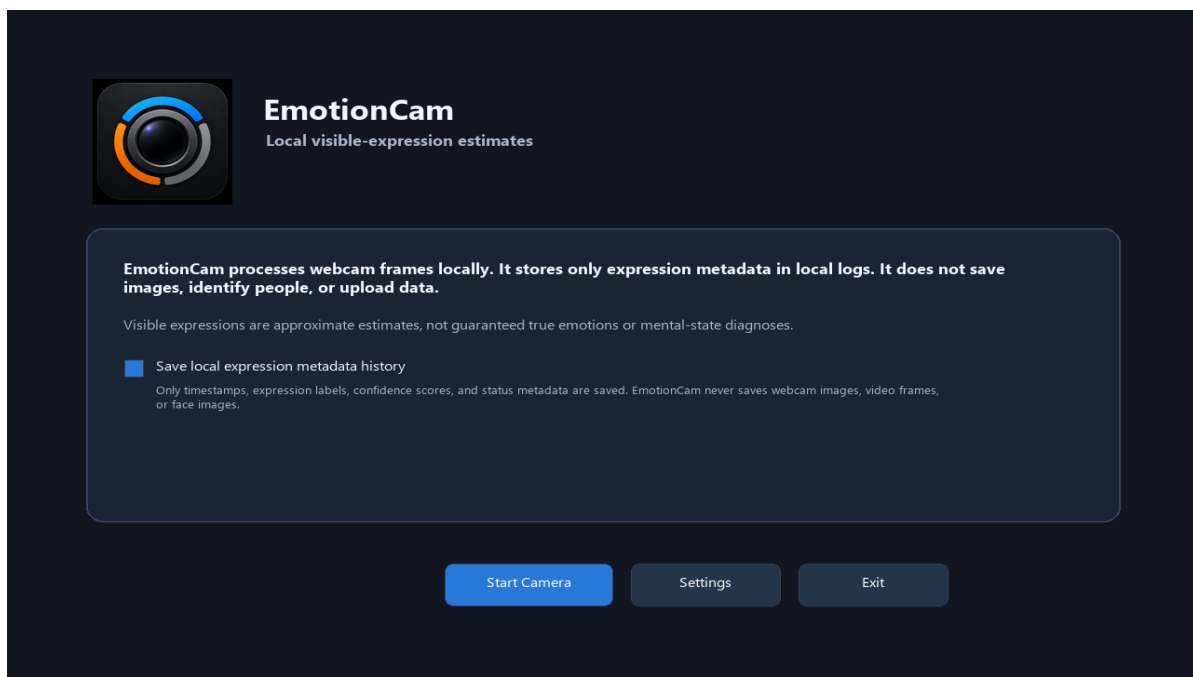
```
python -m pip install --upgrade pip
```

```
python -m pip install -r requirements.txt
```

```
python -m app.main
```

4. First Launch

The startup screen shows the local-processing privacy notice, approximation notice, metadata-history checkbox, Start Camera, Settings, and Exit.

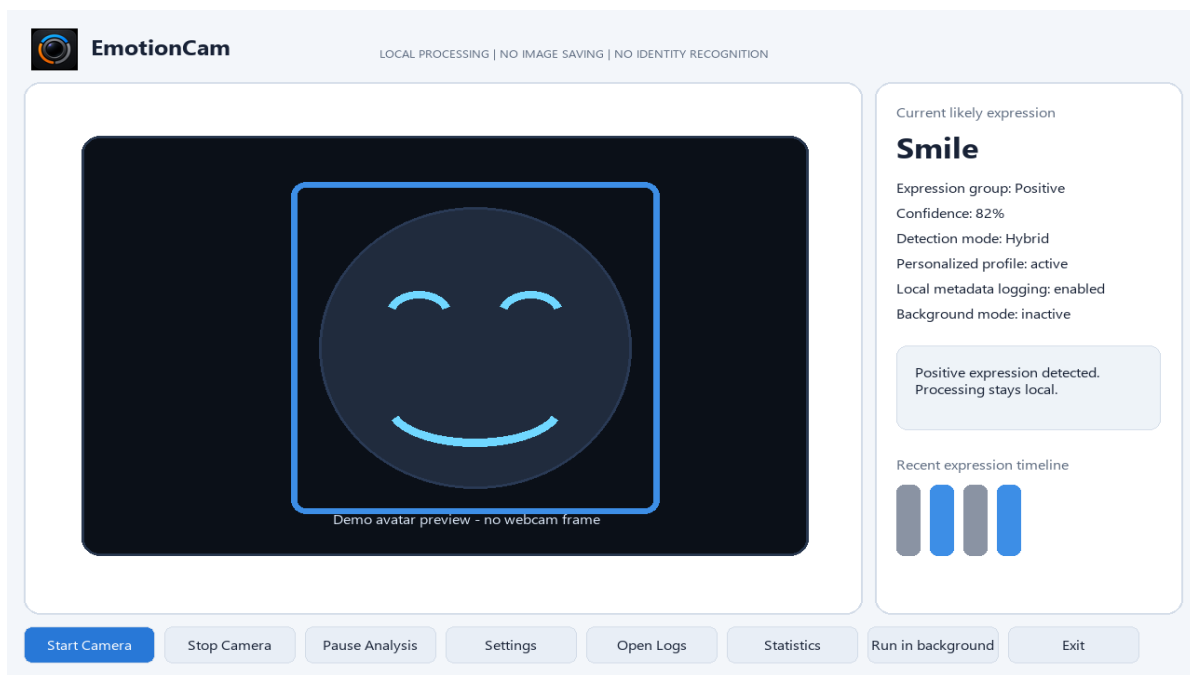
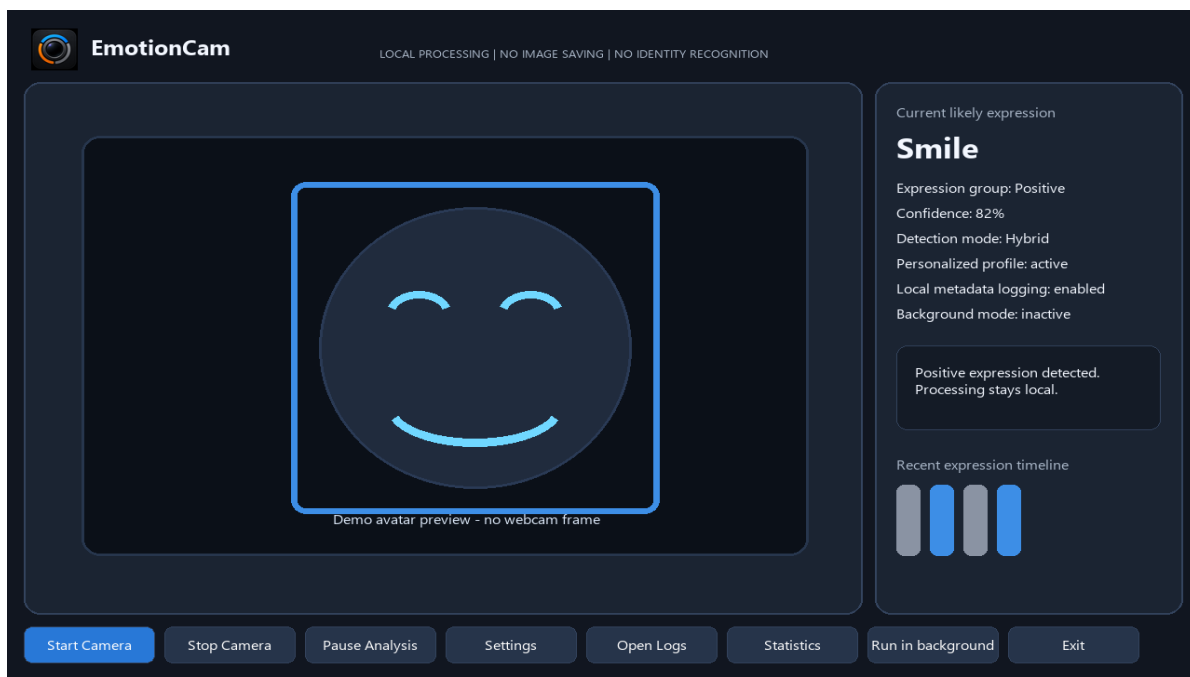


5. Main Dashboard

The dashboard contains the camera preview, face rectangle, current likely expression, expression group, confidence, interaction message, timeline, logging status, privacy status, detection mode, personalized profile status, Statistics, Settings, Open Logs, Run in background, and Exit controls.

Rectangle colors use the smoothed expression group:

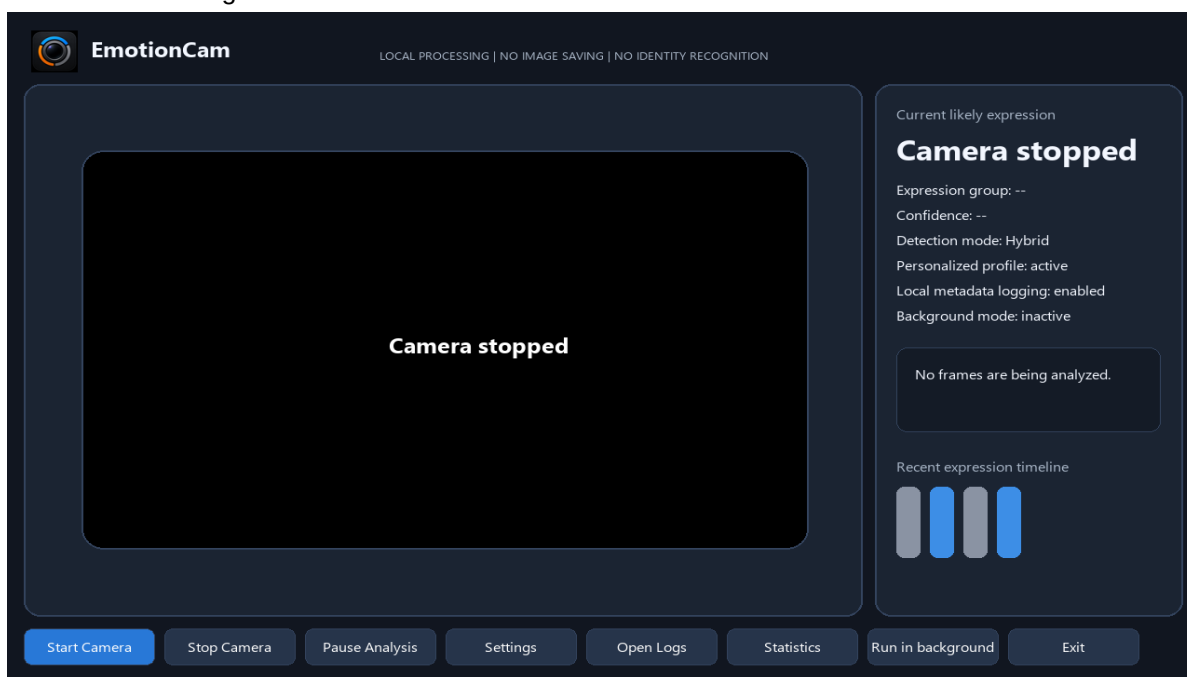
- Blue: positive
- Orange: negative
- Gray: neutral, unknown, low confidence, or no face



6. Starting and Stopping the Camera

- 1 Click Start Camera.
- 1 Wait for the dashboard preview and local analysis status.
- 1 Use Pause Analysis to pause classification while the preview remains active.
- 1 Click Stop Camera to stop analysis, release the camera, and clear the last frame.

- 1 The preview shows a black Camera stopped screen.
- 1 Click Start Camera again to restart.



7. Expression Detection

Heuristic labels include neutral, happy, smile, laughing, surprised, sad, angry, fearful, disgusted, confused, focused, tired, unknown, low_confidence, and no_face.

Personalized calibration also supports smile_small, smile_big, amused, bored, frustrated, concerned, skeptical, thinking, and relaxed.

Group	Labels
Positive	happy, smile, smile_small, smile_big, laughing, amused, surprised
Negative	sad, angry, fearful, disgusted, tired, bored, frustrated, concerned
Neutral	neutral, focused, confused, skeptical, thinking, relaxed, unknown, low_confidence, no_face

Confidence is an estimate. Temporal smoothing and stable-expression timing reduce flicker. Strong negative labels require higher confidence, while uncertain results fall back toward neutral or unknown.

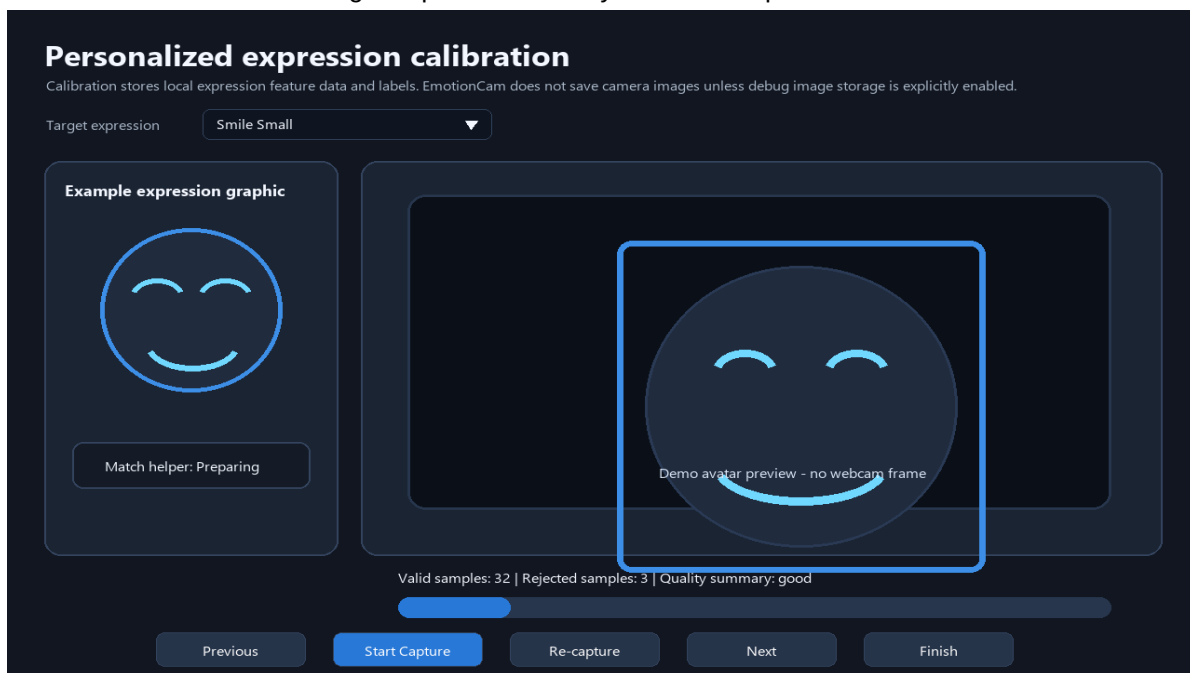
8. Personalized Calibration

Open calibration from Improve detection with calibration or Settings > Train / Calibrate Expressions.

The calibration screen includes:

- Target expression dropdown
- Local example expression graphic
- Demo/live preview area
- Live detected-expression estimate
- Match helper/checkmark
- Previous, Start Capture, Re-capture, Next, and Finish
- Progress, valid sample count, rejected sample count, and quality summary

The old Skip button and separate Label Unknown Expression flow were removed. Users now choose the target expression directly from the dropdown.



Personalized expression calibration

Calibration stores local expression feature data and labels. EmotionCam does not save camera images unless debug image storage is explicitly enabled.

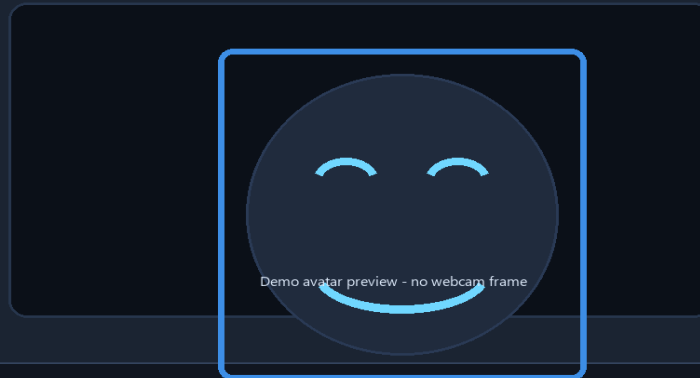
Target expression

Surprised

Example expression graphic



Match helper: Preparing



Valid samples: 32 | Rejected samples: 3 | Quality summary: good

Previous

Start Capture

Re-capture

Next

Finish

Personalized expression calibration

Calibration stores local expression feature data and labels. EmotionCam does not save camera images unless debug image storage is explicitly enabled.

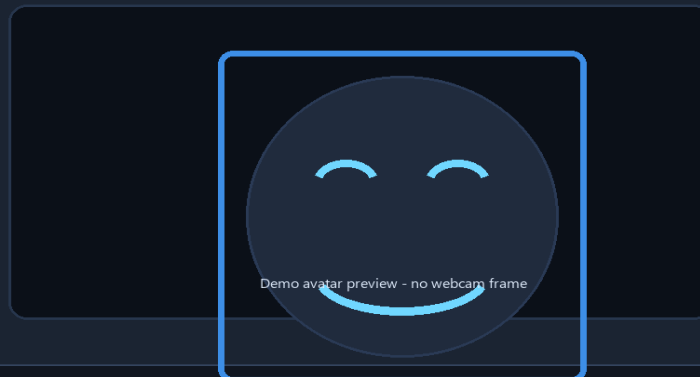
Target expression

Focused

Example expression graphic



Match helper: Preparing



Valid samples: 32 | Rejected samples: 3 | Quality summary: good

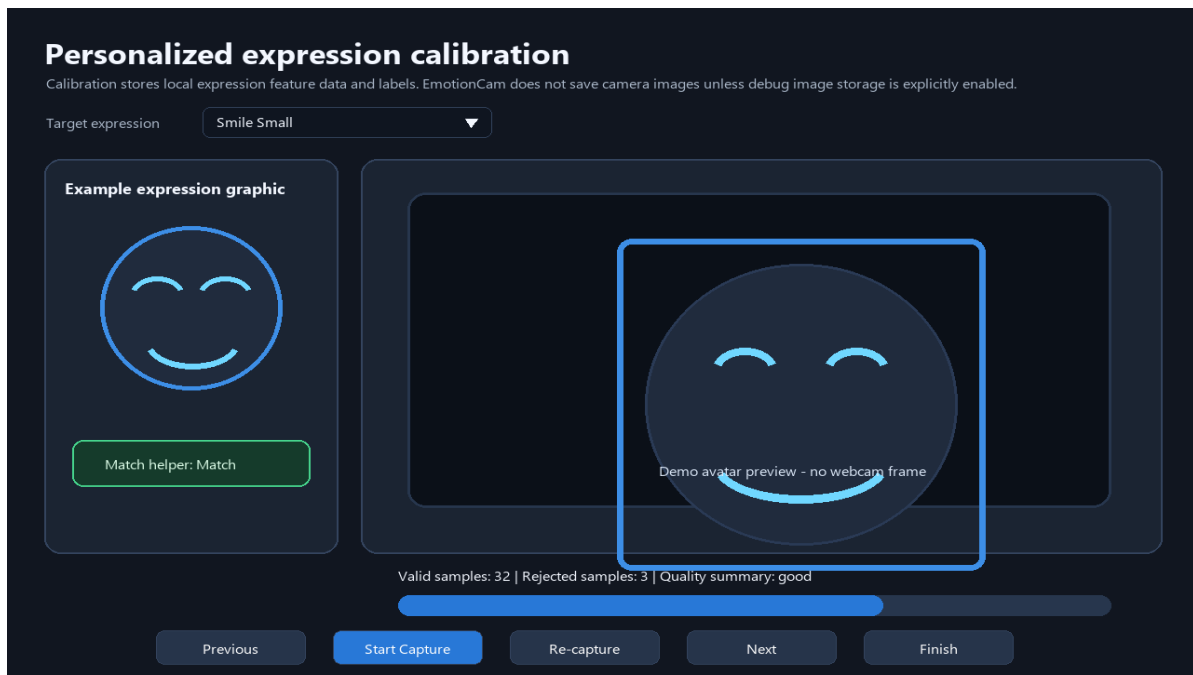
Previous

Start Capture

Re-capture

Next

Finish



9. Settings

Settings include Appearance, User Profile, Daily Email Summary, Camera and Detection, Personalized calibration, Background, Privacy/logging, and Debug options.

Use Appearance > Theme to switch between Dark and Light. The theme is saved locally and applies across the dashboard, Settings, calibration, Statistics, buttons, inputs, progress bars, and message areas.

All numeric fields use spinner controls. You can type values or use the visible up/down arrows.

EmotionCam Settings

Appearance

Theme

Dark / Light

Interface

Polished dashboard, dialogs, controls

User Profile

User name

Optional

Email address

Optional; used only for enabled summaries

Daily Email Summary

Enabled

Off by default

Delivery

Default mail draft or SMTP

Send time

20:00

Camera and Detection

Camera index

0

Detection confidence

0.45

Smoothing window

1.5 s

Face-missing grace

1.0 s

Save

Cancel

EmotionCam Settings

Appearance

Theme

Dark / Light

Interface

Polished dashboard, dialogs, controls

User Profile

User name

Optional

Email address

Optional; used only for enabled summaries

Daily Email Summary

Enabled

Off by default

Delivery

Default mail draft or SMTP

Send time

20:00

Camera and Detection

Camera index

0

Detection confidence

0.45

Smoothing window

1.5 s

Face-missing grace

1.0 s

Save

Cancel

EmotionCam Settings

Camera and Detection

Camera index	0	↕
Detection confidence	0.45	↕
Smoothing window	1.5 s	↕

Appearance

Theme	Dark / Light
Interface	Polished dashboard, dialogs, controls

Save Cancel

EmotionCam Settings

Appearance

Theme	Dark / Light
Interface	Polished dashboard, dialogs, controls

User Profile

User name	Optional
Email address	Optional; used only for enabled summaries

Daily Email Summary

Enabled	Off by default
Delivery	Default mail draft or SMTP
Send time	20:00 ↕

Camera and Detection

Camera index	0	↕
Detection confidence	0.45	↕
Smoothing window	1.5 s	↕
Face-missing grace	1.0 s	↕

Save Cancel

10. Statistics

Click Statistics on the dashboard to open local visual summaries generated from expression metadata logs only.

The Statistics window includes:

- Selected-day positive / neutral / serious-expression balance
- Daily expression-group timeline
- Expression counts by label

- Last-7-days balance
- Most frequent expression
- Approximate analyzed entries/minutes
- Most positive and most serious-expression periods when available
- Refresh and Export Summary controls





11. User Profile and Optional Daily Email

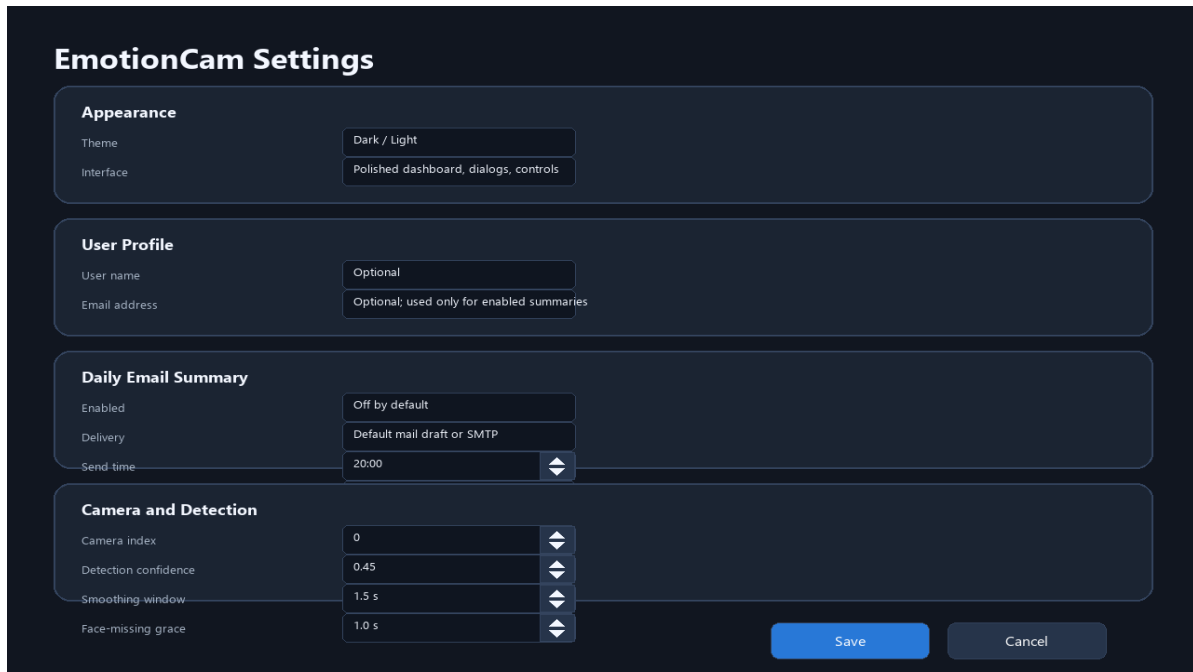
Settings and calibration include optional User name and Email address fields. The name can personalize safe messages. Email is used only for the optional daily summary feature.

Daily email summaries are disabled by default. If enabled, they send local statistics text only. No webcam frames, face images, videos, raw calibration images, or identity data are included.

Delivery modes:

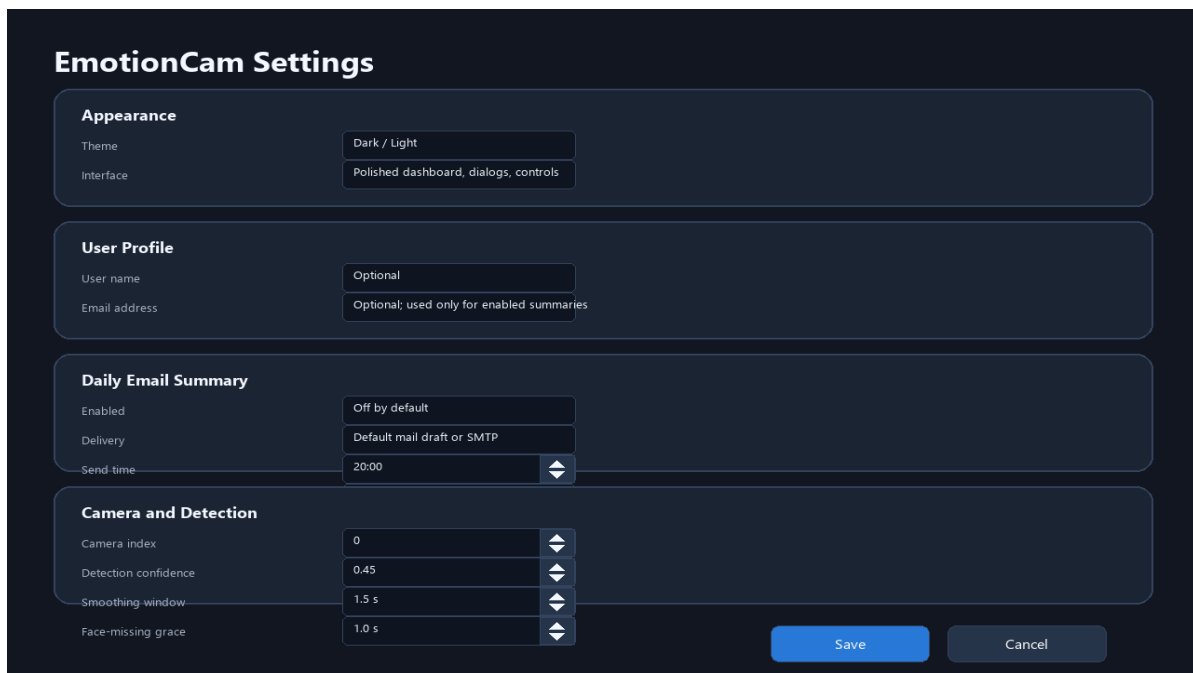
- Default mail client draft: opens a draft for manual review/send.
- SMTP automatic sending: sends at the configured time when the app is running and logs exist.

SMTP passwords are not stored in profile JSON. When Windows credential storage through keyring is available, the password can be stored securely.



The image shows a settings dialog box titled "EmotionCam Settings". It has a dark blue background with white text. The settings are organized into four sections: Appearance, User Profile, Daily Email Summary, and Camera and Detection. Each section contains several settings with input fields or dropdown menus. At the bottom right, there are "Save" and "Cancel" buttons.

Section	Setting	Value
Appearance	Theme	Dark / Light
	Interface	Polished dashboard, dialogs, controls
User Profile	User name	Optional
	Email address	Optional; used only for enabled summaries
Daily Email Summary	Enabled	Off by default
	Delivery	Default mail draft or SMTP
	Send time	20:00
Camera and Detection	Camera index	0
	Detection confidence	0.45
	Smoothing-window	1.5 s
	Face-missing grace	1.0 s



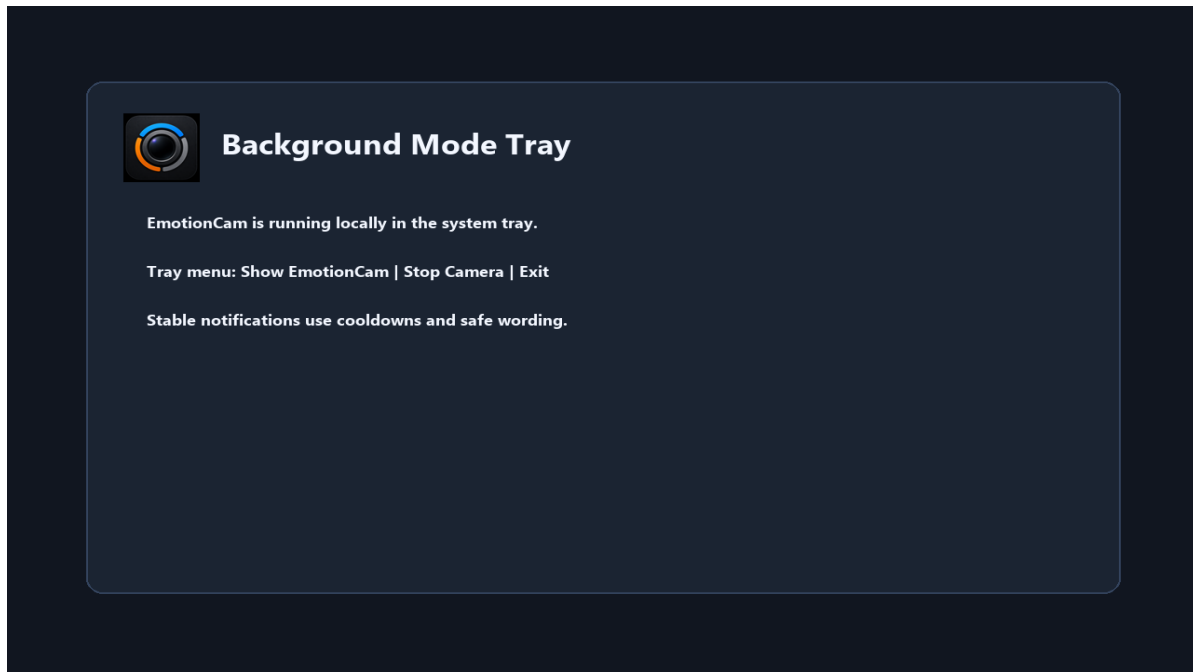
This is a duplicate of the settings dialog box shown above. It contains the same settings and layout.

Section	Setting	Value
Appearance	Theme	Dark / Light
	Interface	Polished dashboard, dialogs, controls
User Profile	User name	Optional
	Email address	Optional; used only for enabled summaries
Daily Email Summary	Enabled	Off by default
	Delivery	Default mail draft or SMTP
	Send time	20:00
Camera and Detection	Camera index	0
	Detection confidence	0.45
	Smoothing-window	1.5 s
	Face-missing grace	1.0 s

12. Background Mode

Click Run in background to keep camera analysis and optional metadata logging active locally while the app hides to the tray when available. The tray menu can show EmotionCam, stop the camera, or exit.

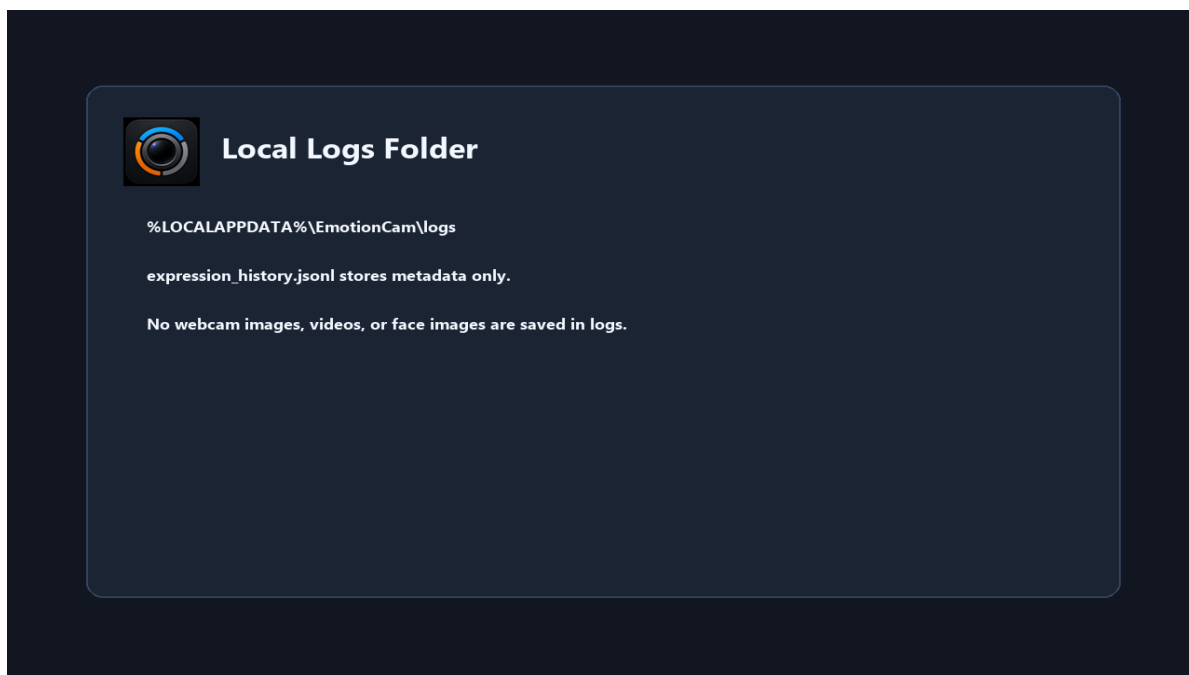
Stable serious-expression estimates can show cooldown-controlled supportive notifications. A later positive shift can show a recovery notification. These are visible-expression estimates, not diagnoses.



13. Logs and Local Data

Data	Path
Settings	%LOCALAPPDATA%\EmotionCam\config.json
Metadata logs	%LOCALAPPDATA%\EmotionCam\logs\expression_history.jsonl
Personalized profile	%LOCALAPPDATA%\EmotionCam\profile\expression_profile.json
User profile and email preferences	%LOCALAPPDATA%\EmotionCam\profile\user_profile.json
Optional debug calibration images	%LOCALAPPDATA%\EmotionCam\profile\debug_images\

Logs are metadata-only and never include webcam frame data.



14. Demo Walkthrough

- 1 Launch EmotionCam and show the icon/startup privacy screen.
- 1 Start the camera and explain visible-expression estimates.
- 1 Show neutral and positive states; point out the blue positive rectangle.
- 1 Stop the camera and show Camera stopped.
- 1 Open Settings and demonstrate spinner arrows.
- 1 Switch Dark to Light and optionally back to Dark.
- 1 Open calibration and choose an expression from the dropdown.
- 1 Show the local example graphic and match helper.
- 1 Use Start Capture, Re-capture, Previous, Next, and Finish.
- 1 Return to the dashboard and show personalized/hybrid status.
- 1 Run in background and explain tray behavior.
- 1 Open Statistics and show local charts.
- 1 Show User Profile and disabled-by-default email summary settings.
- 1 Open Logs and close the app.

For a presenter-ready version, open START_DEMO_HERE.html.

15. Troubleshooting

Issue	What to try
Camera will not start	Allow desktop-app camera access and close other webcam apps.

Issue	What to try
No face detected	Improve lighting, face the camera, move closer, and remove occlusion.
Estimate is wrong	Hold steady, reduce head angle, enable Debug, and calibrate.
Calibration has too few samples	Improve lighting, hold still, or increase capture duration.
Background popups do not show	Enable popups, allow Windows notifications, and wait for cooldown.
Statistics are empty	Enable logging, run analysis briefly, refresh, and select a date with entries.
Daily email is not sent	Confirm opt-in, valid email, SMTP settings/password, logs, send time, and network access.
Installer is blocked	Use SmartScreen **More info > Run anyway** only for a trusted file.

16. Limitations

- Detection is approximate and may be wrong.
- Lighting, blur, occlusion, distance, and head angle affect results.
- Calibration improves estimates but cannot guarantee accuracy.
- The app is designed for one visible user and one laptop webcam.
- Email summaries require network access only after explicit opt-in.

17. Uninstall and Data Removal

Uninstall from Windows Settings > Apps > Installed apps > EmotionCam.

To remove remaining local settings, logs, calibration data, and optional debug frames, fully exit EmotionCam and delete:

%LOCALAPPDATA%\EmotionCam